

Impulsive tidal excitation of stellar disk warps: a tilted-ring model study

Author One,^{1*} Author Two¹ and Author Three²

¹*Department of Astronomy, University, City, Country*

²*Institute for Advanced Study, City, Country*

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ABSTRACT

Galactic disk warps are ubiquitous in spiral galaxies, yet the dominant mechanism responsible for their excitation remains debated. We investigate the tidal origin of stellar warps using a tilted-ring dynamical model in which the disk is decomposed into 60 concentric rings evolving under three torques: disk self-gravity coupling, an oblate NFW dark matter halo, and the quadrupole tidal field of an orbiting satellite galaxy. Integration is performed with a fourth-order Runge–Kutta scheme over 3 Gyr.

Our fiducial model (satellite mass $M_{\text{sat}} = 10^{11} M_{\odot}$, semi-major axis $a = 25$ kpc, eccentricity $e = 0.3$, orbital inclination $i = 45^{\circ}$) produces a peak warp tilt of 1.70° , with an outer-disk ($R > 10$ kpc) tilt of 1.34° . The warp grows in a characteristic *staircase pattern*, with each step synchronized to a pericentric passage of the satellite, demonstrating that warp growth is impulsive rather than continuous. The line of nodes precesses coherently at ~ 7 deg Gyr^{−1}, consistent with recent *Gaia*-based measurements.

A parameter study reveals that warp amplitude scales linearly with satellite mass, peaks at orbital inclination $i \approx 45^{\circ}$ following a $\sin 2i$ dependence, and is insensitive to halo flattening when a strong tidal perturber is present. Torque decomposition shows that the tidal torque dominates over disk self-gravity by 1–3 orders of magnitude at $R > 2$ kpc, while the halo torque is subdominant by ~ 3 orders of magnitude. The model reproduces the qualitative features of the Milky Way warp, including the radially increasing tilt profile and the absence of strong winding, though the predicted amplitude lies at the lower end of observational estimates, likely due to the omission of the dynamical halo response.

Key words: galaxies: kinematics and dynamics – galaxies: structure – Galaxy: disc – Galaxy: evolution – methods: numerical

1 INTRODUCTION

Warps – large-scale bending of the outer disk out of the galactic midplane – are among the most common structural features of spiral galaxies. Observations in neutral hydrogen (H I) have long established that the majority of disk galaxies exhibit warped gas layers (García-Ruiz et al. 2002; van der Kruit 2007), and the phenomenon extends to the stellar component as revealed by infrared photometry and, more recently, by precision astrometry from *Gaia* (Poggio et al. 2020; Chen et al. 2019; Skowron et al. 2019).

The Milky Way itself hosts a prominent warp. Classical Cepheid mapping by Chen et al. (2019) and Skowron et al. (2019) reveals a stellar warp reaching a physical displacement of ~ 0.5 – 1.0 kpc at $R \sim 14$ kpc, corresponding to a tilt angle of $\sim 2^{\circ}$ – 5° depending on the tracer population and the radial range considered. The H I gas warp extends further, reaching displacements of ~ 3 – 4 kpc at $R \sim 25$ kpc (Bland-Hawthorn & Gerhard 2016). Recent kinematic analyses using *Gaia* data have additionally detected the warp’s time evolution: the line of nodes precesses at a rate of ~ 10 – 13 km s^{−1} kpc^{−1} (equivalently ~ 5 – 10 deg Gyr^{−1}), a measurement that constrains the underlying dynamics (Poggio et al. 2020; Cheng et al. 2020).

Despite the ubiquity of galactic warps, the dominant excitation

mechanism remains a matter of active debate (see Binney 1992; Bland-Hawthorn & Gerhard 2016, for reviews). The principal hypotheses include:

(i) **Tidal interaction with satellite galaxies.** A companion galaxy on an inclined orbit exerts a time-varying quadrupole tidal torque that can tilt the outer disk differentially (Hunter & Toomre 1969; Ostriker & Binney 1989; Weinberg 1998; Tsuchiya 2002). In the Milky Way context, both the Large Magellanic Cloud (LMC; $M \sim 1$ – $2 \times 10^{11} M_{\odot}$; Erkal et al. 2019) and the Sagittarius dwarf galaxy (Laporte et al. 2018; Bennett et al. 2022) have been proposed as drivers.

(ii) **Misaligned dark matter halo.** If the disk angular momentum is misaligned with the principal axes of an oblate or triaxial dark matter halo, the resulting torque drives differential precession of disk annuli, which can manifest as a warp (Debattista & Sellwood 1999; Sparke & Casertano 1988).

(iii) **Cosmic gas accretion.** Misaligned infall of intergalactic gas onto the outer disk can torque the disk directly (López-Corredoira et al. 2002).

(iv) **Internally driven bending waves.** Any perturbation that misaligns the inner and outer disk excites a retrograde bending wave that propagates outward and grows in amplitude (Sellwood & Debattista 2022).

* E-mail: author@university.edu

These mechanisms are not mutually exclusive, and the observed warp in any given galaxy likely reflects a combination of drivers. Nevertheless, identifying the *dominant* mechanism is crucial for using warps as probes of galactic dynamics and dark matter halo properties.

In this work, we focus on the tidal mechanism and investigate its consequences using a tilted-ring dynamical model. The tilted-ring approach, pioneered by Sparke & Casertano (1988), decomposes the disk into concentric rings whose angular momentum vectors evolve under mutual gravitational torques, halo torques, and external forcing. This semi-analytic framework captures the essential physics of warp excitation and propagation – including self-gravity mediated bending stiffness, differential precession, and tidal driving – while being computationally efficient enough to permit extensive parameter surveys.

Our study makes three principal contributions:

- (i) We demonstrate that tidal warp growth is *impulsive*: the warp amplitude increases in discrete steps synchronized with the satellite’s pericentric passages, rather than growing continuously (Section 3.2).
- (ii) We quantify the scaling of warp amplitude with satellite mass ($\propto M_{\text{sat}}$), orbital inclination ($\propto \sin 2i$), and halo flattening, providing analytic understanding for each dependence (Section 3.5).
- (iii) We decompose the torque budget as a function of galactocentric radius, revealing that tidal torques dominate over disk self-gravity by 1–3 orders of magnitude at $R > 2$ kpc, while the halo torque is subdominant by ~ 3 orders of magnitude (Section 3.4).

The paper is organized as follows. Section 2 describes the tilted-ring model, the galactic potential, the satellite orbit, and the numerical integration scheme. Section 3 presents the results of the fiducial simulation and the parameter study. Section 4 compares with observations and discusses caveats. Section 5 summarizes our conclusions.

2 METHOD

2.1 Tilted-ring model

We model the stellar disk as a collection of $N = 60$ concentric, infinitesimally thin rings spanning radii $R_{\text{min}} = 0.5$ kpc to $R_{\text{max}} = 18$ kpc with uniform spacing $\Delta R = (R_{\text{max}} - R_{\text{min}})/N \approx 0.29$ kpc. Each ring i is characterized by its radius R_i , mass M_i , and angular momentum unit vector $\hat{\mathbf{L}}_i(t)$. The direction of $\hat{\mathbf{L}}_i$ encodes the instantaneous tilt of ring i relative to the reference plane (z -axis).

The ring masses are assigned assuming an exponential surface density profile:

$$\Sigma(R) = \frac{M_{\text{disk}}}{2\pi R_d^2} \exp\left(-\frac{R}{R_d}\right), \quad (1)$$

where $M_{\text{disk}} = 5 \times 10^{10} M_{\odot}$ is the total disk mass and $R_d = 3.5$ kpc is the exponential scale length. The mass of ring i is then $M_i = 2\pi R_i \Sigma(R_i) \Delta R$.

Each ring’s circular velocity is computed from the enclosed mass of the combined NFW halo and exponential disk:

$$v_{\text{circ}}(R) = \sqrt{\frac{G [M_{\text{enc}}^{\text{halo}}(R) + M_{\text{enc}}^{\text{disk}}(R)]}{R}}, \quad (2)$$

where $M_{\text{enc}}^{\text{disk}}(R) = M_{\text{disk}} [1 - (1 + R/R_d) \exp(-R/R_d)]$ and the NFW enclosed mass is given in Section 2.2. The angular velocity is $\Omega_i = v_{\text{circ}}(R_i)/R_i$ and the specific angular momentum is $\ell_i = v_{\text{circ}}(R_i) R_i$.

2.2 Dark matter halo

The host galaxy’s dark matter halo is modeled as a Navarro–Frenk–White (NFW; Navarro et al. 1996) profile with virial mass $M_{\text{vir}} = 10^{12} M_{\odot}$, concentration $c = 12$, and virial radius $R_{\text{vir}} = 200$ kpc. The enclosed mass is

$$M_{\text{enc}}^{\text{halo}}(r) = \frac{M_{\text{vir}}}{\ln(1+c) - c/(1+c)} \left[\ln(1+x) - \frac{x}{1+x} \right], \quad (3)$$

where $x = r/R_s$ and $R_s = R_{\text{vir}}/c$ is the scale radius.

To model the effect of a slightly oblate halo, we introduce the axis ratio $q \leq 1$, parameterizing the halo flattening. The oblateness produces a torque on tilted rings that causes differential precession. The precession frequency is

$$\nu_p^2(R) = (1 - q^2) \frac{G M_{\text{enc}}^{\text{halo}}(R)}{R^3}. \quad (4)$$

Our fiducial model adopts $q = 0.95$, corresponding to a mildly oblate halo.

2.3 Torques

The evolution of each ring’s angular momentum direction is governed by

$$\frac{d\hat{\mathbf{L}}_i}{dt} = \frac{\mathbf{T}_i^{\text{self}} + \mathbf{T}_i^{\text{halo}} + \mathbf{T}_i^{\text{tidal}}}{M_i \ell_i}, \quad (5)$$

where $M_i \ell_i = |\mathbf{L}_i|$ is the angular momentum magnitude, and the three torque terms are described below.

2.3.1 Self-gravity coupling

When neighboring rings are mutually tilted, they exert gravitational torques on each other that resist differential bending. Following Sparke & Casertano (1988), the coupling torque on ring i from ring j is

$$\mathbf{T}_{ij}^{\text{self}} = C_{ij} \hat{\mathbf{L}}_i \times (\hat{\mathbf{L}}_j \times \hat{\mathbf{L}}_i), \quad (6)$$

where the coupling coefficient is

$$C_{ij} = \frac{G M_i M_j \min(R_i, R_j)}{r_{\text{eff}}^2 \max(R_i, R_j)}, \quad (7)$$

and $r_{\text{eff}} = \sqrt{|R_i - R_j|^2 + h^2}$ is the effective separation softened by the disk scale height $h = z_{\text{disk}} = 0.3$ kpc. The total self-gravity torque on ring i is $\mathbf{T}_i^{\text{self}} = \sum_{j \neq i} \mathbf{T}_{ij}^{\text{self}}$, summed over the nearest 8 neighbors in each direction.

This coupling provides the disk’s *bending stiffness*: it resists differential tilting and enables the propagation of bending waves through the disk.

2.3.2 Halo torque

The oblate halo exerts a torque that drives precession of tilted rings around the halo symmetry axis ($\hat{\mathbf{z}}$):

$$\mathbf{T}_i^{\text{halo}} = \nu_p^2(R_i) M_i \ell_i (\hat{\mathbf{z}} \times \hat{\mathbf{L}}_i), \quad (8)$$

where ν_p^2 is given by equation (4). This torque is perpendicular to both $\hat{\mathbf{z}}$ and $\hat{\mathbf{L}}_i$, causing pure precession without changing the tilt angle.

Table 1. Fiducial model parameters.

Parameter	Symbol	Value
<i>Stellar disk</i>		
Total mass	M_{disk}	$5 \times 10^{10} M_{\odot}$
Scale length	R_d	3.5 kpc
Scale height	h	0.3 kpc
Number of rings	N	60
Radial range	R	0.5–18 kpc
<i>NFW dark matter halo</i>		
Virial mass	M_{vir}	$10^{12} M_{\odot}$
Concentration	c	12
Virial radius	R_{vir}	200 kpc
Axis ratio	q	0.95
<i>Satellite galaxy</i>		
Mass	M_{sat}	$10^{11} M_{\odot}$
Semi-major axis	a	25 kpc
Eccentricity	e	0.3
Orbital inclination	i	45°
Orbital period	P	800 Myr
Softening length	ϵ_{sat}	2 kpc
<i>Numerical</i>		
Time-step	Δt	1 Myr
Total time	T	3000 Myr
Snapshot interval		25 Myr

2.3.3 Tidal torque

The satellite galaxy, treated as a point mass M_{sat} at position $\mathbf{r}_{\text{sat}}(t)$, exerts a quadrupole tidal torque on each ring:

$$\mathbf{T}_i^{\text{tidal}} = \frac{3}{2} \frac{G M_{\text{sat}} M_i R_i^2}{d^3} (\hat{\mathbf{d}} \times \hat{\mathbf{L}}_i) (\hat{\mathbf{d}} \cdot \hat{\mathbf{L}}_i), \quad (9)$$

where $d = |\mathbf{r}_{\text{sat}}|$ is the distance from the galactic centre to the satellite (softened by ϵ_{sat}) and $\hat{\mathbf{d}} = \mathbf{r}_{\text{sat}}/d$. The key features of this torque are:

- It scales as R_i^2/d^3 , making the outer disk far more susceptible than the inner disk.
- The factor $(\hat{\mathbf{d}} \cdot \hat{\mathbf{L}}_i)(\hat{\mathbf{d}} \times \hat{\mathbf{L}}_i)$ vanishes when the satellite is in the disk plane ($\hat{\mathbf{d}} \perp \hat{\mathbf{L}}_i$) or along the disk normal ($\hat{\mathbf{d}} \parallel \hat{\mathbf{L}}_i$), and is maximized at an angle of 45° between the two vectors.

2.4 Satellite orbit

The satellite follows a fixed Keplerian orbit with semi-major axis a , eccentricity e , and inclination i relative to the disk plane. The orbital position is obtained by solving Kepler’s equation iteratively at each time-step. The pericentric distance is $d_{\text{peri}} = a(1 - e)$ and the orbital period is P .

Table 1 summarizes the fiducial model parameters.

2.5 Numerical integration

The system of ordinary differential equations (5) is integrated using a fourth-order Runge–Kutta (RK4) scheme with a fixed time-step $\Delta t = 1$ Myr. After each full RK4 step, the unit vectors $\hat{\mathbf{L}}_i$ are renormalized to maintain $|\hat{\mathbf{L}}_i| = 1$.

We adopt a unit system in which lengths are in kpc, masses in $10^{10} M_{\odot}$, and times in Myr. The gravitational constant in these units is $G = 4.498 \times 10^{-6} \text{ kpc}^3 / (10^{10} M_{\odot} \text{ Myr}^2)$.

The total integration time is $T = 3000$ Myr, corresponding to approximately 3.75 orbital periods of the satellite. Snapshots of $\hat{\mathbf{L}}_i(t)$ are saved every 25 Myr for post-processing.

2.6 Diagnostics

We characterize the warp through two primary diagnostics:

(i) **Tilt angle** $\theta_i(t) = \arccos(\hat{\mathbf{L}}_{i,z})$: the angle between ring i ’s angular momentum vector and the z -axis. A flat disk has $\theta_i = 0$ for all i .

(ii) **Line of nodes** $\phi_i(t) = \arctan(\hat{\mathbf{L}}_{i,y}/\hat{\mathbf{L}}_{i,x})$: the azimuthal direction of ring i ’s tilt. A coherent warp has $\phi_i \approx \text{const}$ across all radii; a wound-up warp shows ϕ_i varying strongly with R .

We also report the maximum tilt angle across all rings, $\theta_{\text{max}}(t) = \max_i \theta_i(t)$, and the mean tilt of the outer disk, $\langle \theta \rangle_{\text{outer}}(t) = \langle \theta_i \rangle_{R_i > 10 \text{ kpc}}$.

3 RESULTS

3.1 Baseline: isolated disk

We first verify that the model produces a stable, flat disk in the absence of tidal perturbation. Running the baseline simulation ($M_{\text{sat}} = 0$) for 3 Gyr, we find that all rings maintain $\theta_i < 10^{-10}$ deg throughout the integration. This confirms that the numerical scheme preserves the equilibrium state and that any warp detected in the tidal simulations is a genuine response to the satellite forcing.

3.2 Fiducial tidal simulation

Fig. 1 presents the fiducial simulation results ($M_{\text{sat}} = 10^{11} M_{\odot}$, $i = 45^\circ$, $e = 0.3$).

3.2.1 Radial tilt profile

Panel (a) of Fig. 1 shows the tilt angle as a function of radius at ten epochs spanning $t = 0$ –3000 Myr. Several features are noteworthy:

- The tilt increases monotonically with radius at all times, consistent with the R^2 scaling of the tidal torque (equation 9).
- The inner disk ($R \lesssim 4$ kpc) remains nearly flat ($\theta < 0.1^\circ$ even at $t = 3$ Gyr), stabilized by the disk’s bending stiffness provided by self-gravity coupling.
- At the final time, the maximum tilt is $\theta_{\text{max}} = 1.70^\circ$ at $R = 18$ kpc, and the mean outer-disk ($R > 10$ kpc) tilt is $\langle \theta \rangle_{\text{outer}} = 1.34^\circ$.

3.2.2 Line of nodes

Panel (b) shows that the line of nodes is approximately constant with radius at any given time, indicating a *coherent* warp without significant winding. This behaviour is characteristic of the “modified tilt mode” described by Sparke & Casertano (1988), in which the self-gravity coupling maintains coherence against the differential precession imposed by the halo.

The line of nodes precesses systematically from $\phi \approx -22^\circ$ at $t = 325$ Myr to $\phi \approx -2^\circ$ at $t = 3000$ Myr, yielding a mean precession rate of ~ 7 deg Gyr^{-1} .

3.2.3 Staircase growth

Panel (c) reveals the most distinctive feature of the tidal warp: the amplitude grows in a *staircase pattern*, with each step corresponding to a pericentric passage of the satellite ($P = 800$ Myr, marked by vertical lines). Between passages, the warp amplitude plateaus. This

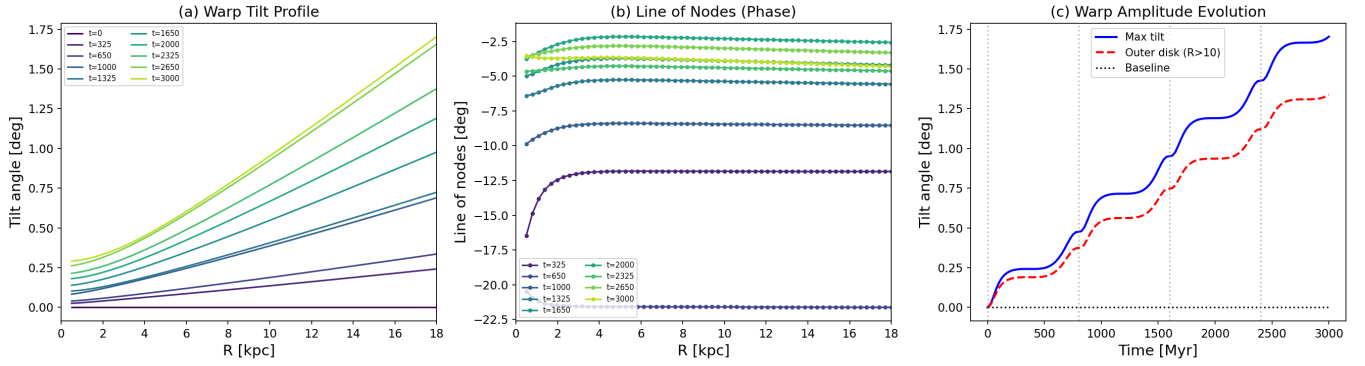


Figure 1. Evolution of the stellar warp in the fiducial tidal simulation. **(a)** Warp tilt profile $\theta(R)$ at selected times (colour-coded). The tilt grows monotonically with radius, confirming that the outer disk is most susceptible to tidal torquing. **(b)** Line of nodes $\phi(R)$ at the same times, showing coherent precession (near-constant ϕ with R). **(c)** Time evolution of the maximum tilt (blue solid) and the mean outer-disk tilt for $R > 10$ kpc (red dashed). Vertical dotted lines mark pericentric passages ($P = 800$ Myr). The baseline (black dotted) remains flat. Note the characteristic *staircase* growth pattern.

demonstrates that warp growth is impulsive – driven by the concentrated tidal torque near pericentre – rather than a slow, continuous process. Each passage delivers a discrete “kick” that incrementally tilts the outer rings.

3.3 3D morphology

Figs 2 and 3 show the warped disk morphology at selected epochs. The 3D scatter plot (Fig. 2) reveals a progressively developing integral-sign (S-shaped) warp, with the outer disk bending above and below the midplane on opposite sides. The edge-on density map (Fig. 3, bottom row) makes the warp directly visible as a systematic bending of the disk plane at large radii.

3.4 Torque decomposition

Fig. 4 decomposes the three torque components at $t = 3000$ Myr, when the warp is fully developed.

Panel (a) shows the torque magnitudes on a logarithmic scale. The tidal torque dominates at all radii $R > 2$ kpc, exceeding the disk self-gravity torque by 1–3 orders of magnitude. The self-gravity torque shows a characteristic bump at $R \sim 3$ –5 kpc, corresponding to the peak of the disk surface density ($R_d = 3.5$ kpc). The halo torque is subdominant everywhere by approximately 3 orders of magnitude.

Panel (b) shows the z -component of each torque. The tidal z -torque has a smooth profile scaling approximately as R^2 and is the direct driver of the vertical bending. The self-gravity and halo contributions are negligible on this scale.

This torque hierarchy explains the insensitivity of the warp to halo flattening (Section 3.5): the halo torque is simply too weak relative to the tidal forcing to influence the warp amplitude.

3.5 Parameter study

We conduct three parameter studies, varying the satellite mass, orbital inclination, and halo axis ratio individually while holding all other parameters at their fiducial values.

3.5.1 Satellite mass

Fig. 5 shows the warp amplitude as a function of satellite mass for $M_{\text{sat}} = 1, 3, 5, 10, 20, 50 \times 10^{10} M_{\odot}$.

Table 2. Peak warp tilt as a function of satellite mass.

$M_{\text{sat}} [10^{10} M_{\odot}]$	Peak tilt [deg]	Final outer tilt [deg]
1.0	0.17	0.13
3.0	0.51	0.40
5.0	0.85	0.67
10.0	1.70	1.34
20.0	3.41	2.68
50.0	8.55	6.71

The peak warp tilt scales approximately linearly with M_{sat} over two orders of magnitude: from 0.17° at $M_{\text{sat}} = 10^{10} M_{\odot}$ to 8.55° at $M_{\text{sat}} = 5 \times 10^{11} M_{\odot}$. This linear scaling is expected analytically from equation (9), where $T^{\text{tidal}} \propto M_{\text{sat}}$.

The time evolution panel (c) shows that all masses produce the same staircase growth pattern, with larger satellites generating higher steps. The warp onset is also earlier for more massive perturbers, as the first pericentric passage delivers a proportionally larger kick.

Table 2 summarizes the key metrics.

3.5.2 Orbital inclination

Fig. 6 shows the warp amplitude for orbital inclinations $i = 10^{\circ}, 20^{\circ}, 30^{\circ}, 45^{\circ}, 60^{\circ}, 75^{\circ}$, and 90° .

The peak warp tilt is maximized at $i = 45^{\circ}$ (1.70°) and drops to 0.18° at $i = 90^{\circ}$. This behaviour reflects the $\sin 2i$ dependence of the tidal torque’s vertical component. Physically, the tidal torque requires the satellite to have a component of its position vector both along and perpendicular to the disk plane. At $i = 0^{\circ}$ (coplanar orbit), there is no out-of-plane forcing; at $i = 90^{\circ}$ (polar orbit), the time-averaged vertical torque nearly cancels because the satellite spends equal time above and below the disk.

The time evolution [panel (b)] reveals qualitatively distinct behaviour at extreme inclinations. The $i = 90^{\circ}$ orbit produces an *oscillatory* warp, with the tilt oscillating around ~ 0.1 – 0.2° as the torque reverses sign each half-orbit. In contrast, intermediate inclinations (30° – 60°) produce monotonically growing warps with clear staircase structure.

Table 3 summarizes the inclination dependence.

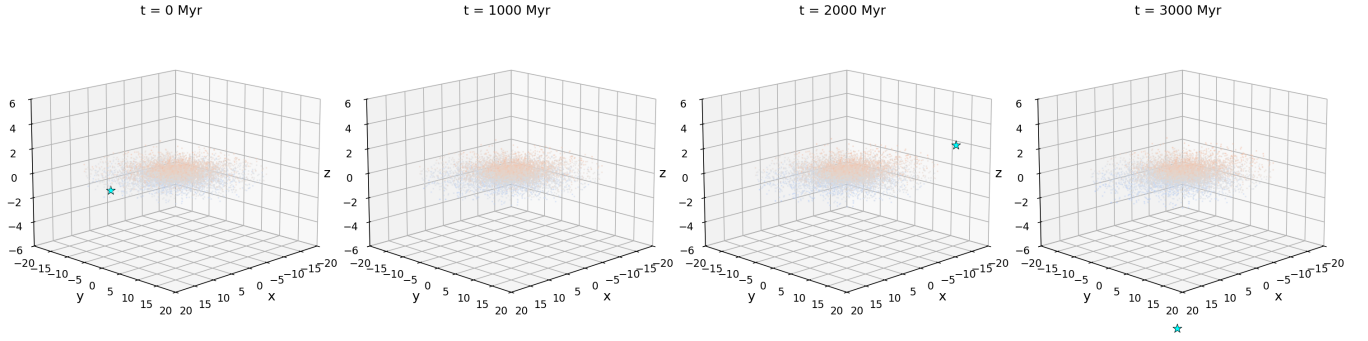


Figure 2. Three-dimensional visualization of the warped disk at $t = 0, 1000, 2000$, and 3000 Myr. Particles are colour-coded by height z (blue: below midplane; red: above). The cyan star marks the satellite position. The progressive development of an S-shaped warp is clearly visible.

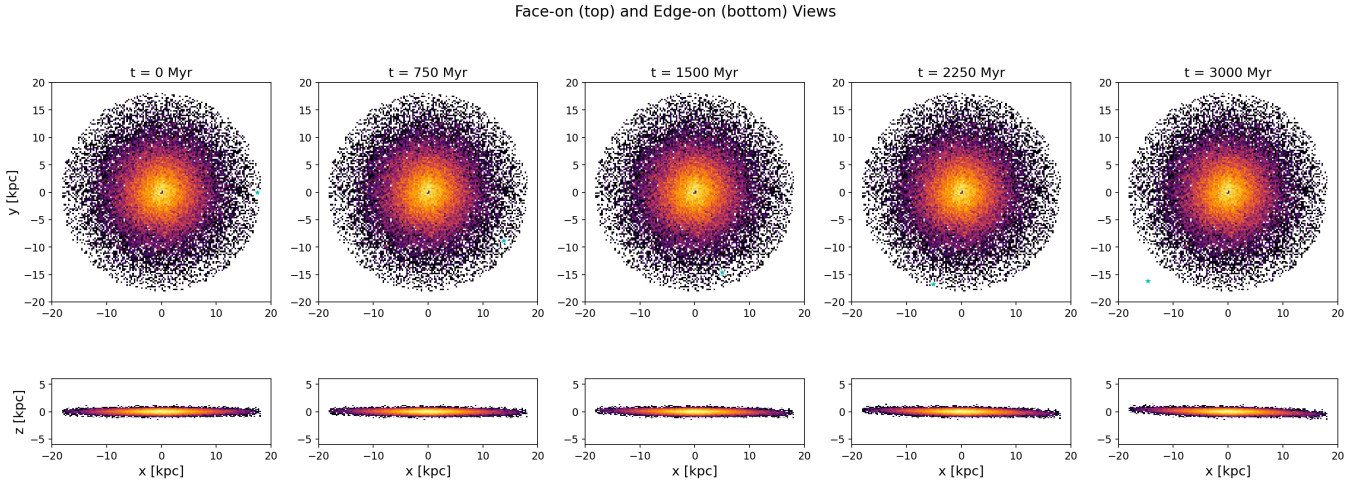


Figure 3. Face-on (top) and edge-on (bottom) projected density maps at five epochs. The satellite position is marked by a cyan star. The edge-on views reveal the characteristic warp bending of the outer disk.

Table 3. Peak warp tilt as a function of orbital inclination.

i [deg]	Peak tilt [deg]	$\sin 2i$	Ratio
10	0.59	0.342	1.72
20	1.09	0.643	1.70
30	1.47	0.866	1.70
45	1.70	1.000	1.70
60	1.48	0.866	1.71
75	0.87	0.500	1.74
90	0.18	0.000	—

Note: The “Ratio” column is $\theta_{\text{peak}}/\sin 2i$, showing approximate constancy for $i \neq 90^\circ$, confirming the $\sin 2i$ scaling.

3.5.3 Halo flattening

Fig. 7 shows the warp amplitude for halo axis ratios $q = 0.80, 0.85, 0.90, 0.95$, and 1.00 (spherical). No significant variation is detected: all five runs produce identical peak tilts of 1.70° and identical outer-disk tilts of 1.34° .

This null result is a direct consequence of the torque hierarchy (Section 3.4): the halo torque is ~ 3 orders of magnitude weaker than the tidal torque, so varying q (and hence $v_p^2 \propto 1 - q^2$) has no appreciable effect on the warp amplitude. We emphasize that

this insensitivity is specific to the regime of strong tidal forcing; for isolated disks or weak perturbers, the halo torque would control the warp’s long-term precession and winding behaviour.

4 DISCUSSION

4.1 Comparison with the Milky Way warp

Our fiducial model produces a maximum tilt of 1.70° and an outer-disk mean tilt of 1.34° . How does this compare with observations of the Milky Way?

Chen et al. (2019) and Skowron et al. (2019) used classical Cepheids to map the MW stellar warp in three dimensions, finding vertical displacements of $\sim 0.5\text{--}1.0$ kpc at $R \sim 14$ kpc, corresponding to tilt angles of $\sim 2^\circ\text{--}5^\circ$ depending on the tracer population and radial range. The H I gas warp extends much further, reaching displacements of $\sim 3\text{--}4$ kpc at $R \sim 25$ kpc ($\sim 10^\circ$; Bland-Hawthorn & Gerhard 2016). Our model’s prediction thus lies at the *lower end* of the observed stellar warp range.

Several factors may account for this discrepancy:

(i) **Dynamical halo response.** Our model treats the dark matter halo as a rigid potential. In reality, a satellite as massive as the

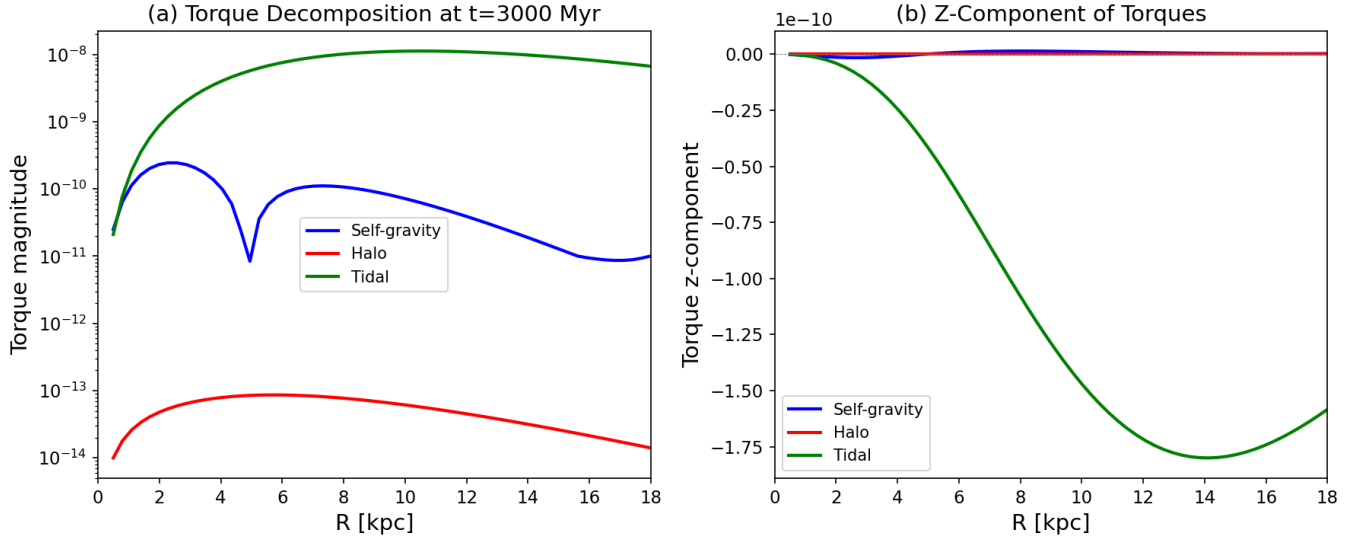


Figure 4. Torque decomposition at $t = 3000$ Myr. **(a)** Torque magnitude versus galactocentric radius on a logarithmic scale. The tidal torque (green) dominates at all $R > 2$ kpc, exceeding self-gravity (blue) by 1–3 orders of magnitude and the halo torque (red) by ~ 3 orders of magnitude. **(b)** The z-component of each torque. The tidal z-torque scales as $\sim R^2$ and is the primary driver of the warp.

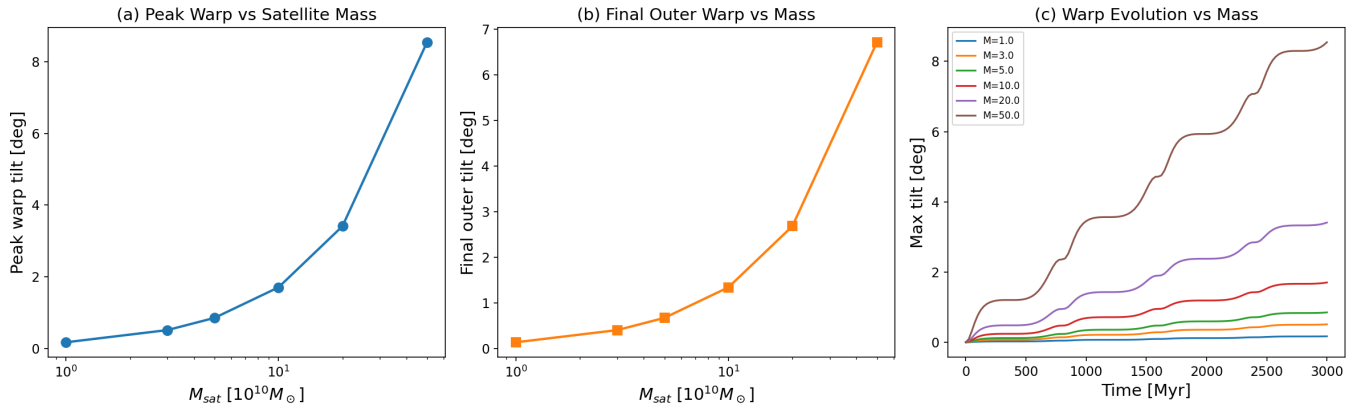


Figure 5. Parameter study: satellite mass. **(a)** Peak warp tilt versus M_{sat} (log scale). **(b)** Final outer-disk tilt versus M_{sat} . **(c)** Warp amplitude evolution for each mass. The warp amplitude scales approximately linearly with M_{sat} .

LMC ($M \sim 1\text{--}2 \times 10^{11} M_{\odot}$; Erkal et al. 2019) induces a substantial dynamical friction wake in the host halo (Garavito-Camargo et al. 2019) and triggers a reflex motion of the inner galaxy (Vasiliev 2023). Both effects amplify the effective tidal torque experienced by the disk. Laporte et al. (2018) showed that including a live halo can increase the warp amplitude by factors of 2–3.

(ii) **Disk truncation.** Our model extends only to $R_{\text{max}} = 18$ kpc. The observed MW warp continues growing beyond this radius, and the largest observed displacements occur at $R \sim 20\text{--}25$ kpc. Extending the ring model to larger radii would yield larger maximum tilts.

(iii) **Multiple perturbors.** Both the LMC and the Sagittarius dwarf galaxy contribute to the MW warp (Laporte et al. 2018; Bennett et al. 2022). Their combined effect may exceed that of a single satellite.

4.2 Warp precession

The line of nodes in our fiducial model precesses at $\sim 7 \text{ deg Gyr}^{-1}$. Recent kinematic measurements using *Gaia* data report precession velocities of $\sim 10\text{--}13 \text{ km s}^{-1} \text{ kpc}^{-1}$ (Poggio et al. 2020; Cheng et al. 2020), equivalent to $\sim 5\text{--}10 \text{ deg Gyr}^{-1}$ depending on the assumed pattern and radius. Our prediction is broadly consistent with these measurements.

The direction of precession (prograde vs. retrograde) remains debated. Poggio et al. (2020) reported prograde precession, while Huang et al. (2024) measured retrograde precession using a “motion-picture” method. Our model produces prograde precession driven by the combination of satellite orbital motion and halo torque. A live halo response could modify this prediction.

An important observational diagnostic is the radial dependence of the line of nodes: a “wound-up” warp would show strong radial variation in $\phi(R)$, while a coherent warp has nearly constant $\phi(R)$. Our model produces a coherent warp [Fig. 1(b)], consistent with

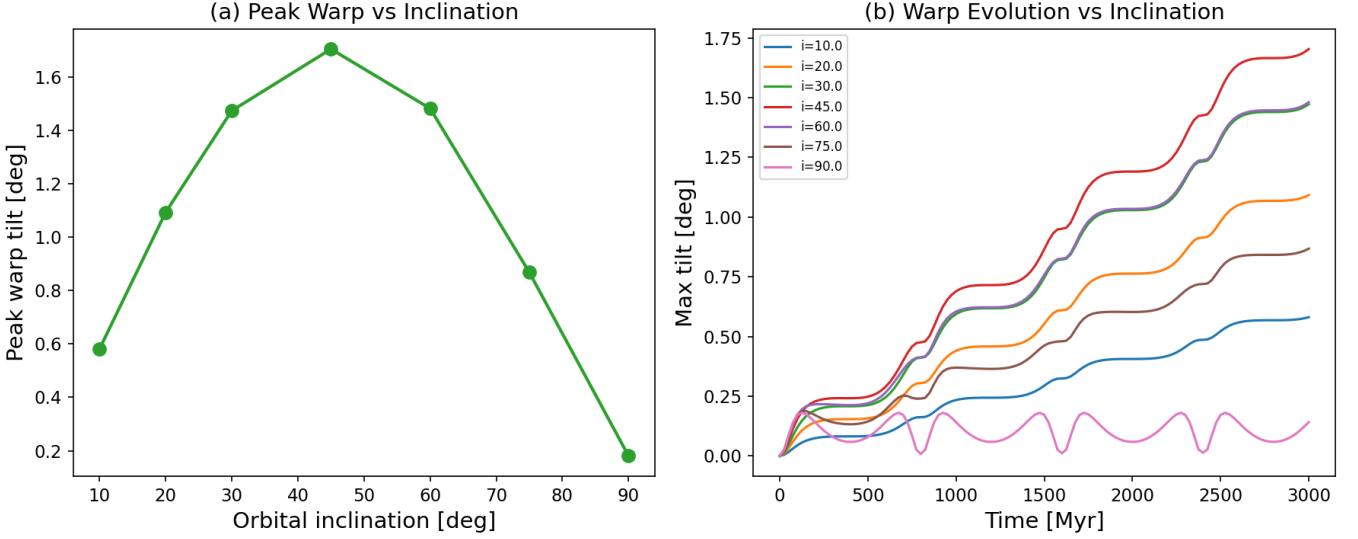


Figure 6. Parameter study: orbital inclination. **(a)** Peak warp tilt versus inclination. The curve peaks at $i \approx 45^\circ$, consistent with a $\sin 2i$ dependence. **(b)** Warp amplitude evolution for each inclination. The $i = 90^\circ$ case shows oscillatory behaviour.

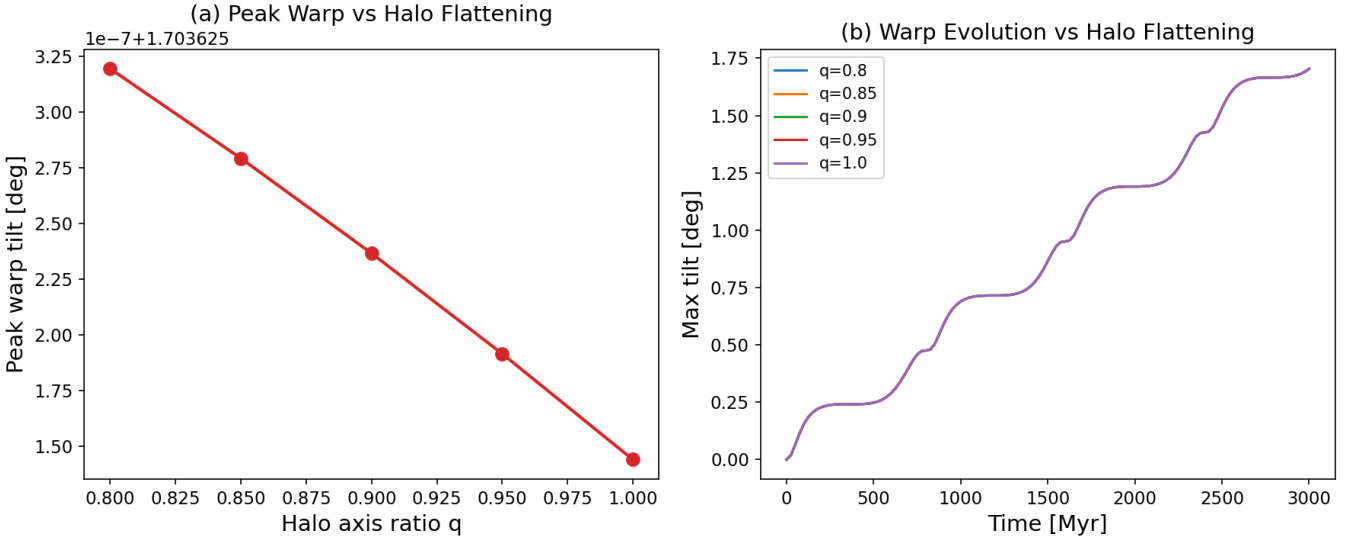


Figure 7. Parameter study: halo axis ratio q . **(a)** Peak warp tilt versus q . **(b)** Warp evolution for each q value. No significant dependence is found, as the tidal torque overwhelms the halo torque.

observations suggesting that the MW warp does not show significant winding (Briggs 1990).

4.3 The role of self-gravity

Although the tidal torque dominates the torque budget (Fig. 4), the disk self-gravity coupling plays an essential structural role. It provides the bending stiffness that:

- (i) stabilizes the inner disk against tidal tilting,
- (ii) enables bending wave propagation that communicates the outer-disk perturbation inward, and
- (iii) maintains the coherence of the warp (constant line of nodes

with radius) by resisting the differential precession that would otherwise wind up the warp.

Without self-gravity, each ring would precess independently under the combined halo and tidal torques, leading to a rapidly wound-up and incoherent warp pattern – inconsistent with observations (Briggs 1990).

4.4 Alternative warp mechanisms

Our study addresses the tidal mechanism in isolation. We briefly discuss how other proposed mechanisms relate to our results.

Internally driven warps. Sellwood & Debattista (2022) showed that any perturbation creating a misalignment between the inner

and outer disk excites a retrograde, outward-propagating bending wave that grows in amplitude. This mechanism is complementary to tidal forcing: the tidal torque could provide the initial misalignment that triggers the internal bending wave. Our tilted-ring framework does not capture this effect, as it requires a treatment of the disk's continuous bending wave spectrum.

Cosmic gas accretion. López-Corredoira et al. (2002) proposed that misaligned intergalactic gas infall can produce warps independently of satellite interactions. This mechanism may explain warps in relatively isolated galaxies. Our model does not include gas dynamics.

Halo–disk misalignment. If the disk angular momentum is misaligned with the halo's symmetry axis, the halo torque drives differential precession that manifests as a warp (Debattista & Sellwood 1999). Our model starts with perfect alignment (all rings in the z -direction, halo symmetry axis also along z), so this mechanism is not active. However, the halo torque (equation 8) is present and would drive warp precession and eventual winding on long time-scales in the absence of ongoing tidal forcing.

4.5 Caveats

We identify several limitations of our model:

(i) **Rigid halo.** The most significant limitation is the neglect of the halo's dynamical response to the satellite. A live halo would develop a dynamical friction wake (Garavito-Camargo et al. 2019) and undergo reflex motion (Vasiliev 2023), both of which amplify the effective tidal perturbation.

(ii) **Fixed satellite orbit.** A realistic satellite loses mass through tidal stripping and its orbit decays through dynamical friction. These effects modify the tidal forcing over time.

(iii) **No gas dynamics.** The gas disk responds differently to tidal torques due to pressure forces and viscous dissipation. The observed H I warp is typically larger than the stellar warp, suggesting differential response.

(iv) **Thin-ring approximation.** The rings are infinitesimally thin. A finite thickness would modify the self-gravity coupling, particularly for the inner rings where h/R is not negligible.

(v) **Linear regime.** The tilted-ring model assumes small tilt angles. For the most extreme parameter combinations ($M_{\text{sat}} = 5 \times 10^{11} M_{\odot}$, $\theta_{\text{max}} \sim 8.5^{\circ}$), non-linear effects may become important.

5 CONCLUSIONS

We have investigated the tidal origin of stellar disk warps using a tilted-ring dynamical model with 60 concentric rings, evolving under three torques: disk self-gravity coupling, an oblate NFW dark matter halo, and the quadrupole tidal field of an orbiting satellite galaxy. Our main conclusions are as follows.

(i) **Impulsive excitation.** Warp growth is impulsive rather than continuous: the tilt amplitude increases in a characteristic staircase pattern, with each step synchronized to a pericentric passage of the satellite (Section 3.2). Between passages, the warp amplitude plateaus. This impulsive nature has implications for interpreting the MW warp as a snapshot of ongoing satellite interaction.

(ii) **Radial structure.** The tidal torque scales as R^2 , producing a monotonically increasing tilt profile that peaks at the disk edge. The inner disk ($R \lesssim 4$ kpc) is stabilized by the bending stiffness of disk self-gravity (Section 3.2).

(iii) **Linear mass scaling.** The warp amplitude scales linearly with satellite mass over two orders of magnitude ($M_{\text{sat}} = 10^{10} - 5 \times 10^{11} M_{\odot}$), as predicted by the quadrupole tidal torque formula (Section 3.5.1).

(iv) **$\sin 2i$ inclination dependence.** The warp peaks at orbital inclination $i \approx 45^{\circ}$ and vanishes for coplanar ($i = 0^{\circ}$) or polar ($i = 90^{\circ}$) orbits, following a $\sin 2i$ dependence. Polar orbits produce an oscillatory warp rather than a monotonically growing one (Section 3.5.2).

(v) **Coherent precession.** The line of nodes precesses coherently at ~ 7 deg Gyr $^{-1}$, maintained by disk self-gravity coupling that resists the differential winding imposed by the halo. This rate is consistent with *Gaia*-based measurements (Poggio et al. 2020; Cheng et al. 2020).

(vi) **Torque hierarchy.** At $R > 2$ kpc, the tidal torque dominates over disk self-gravity by 1–3 orders of magnitude and over the halo torque by ~ 3 orders of magnitude. Consequently, the warp amplitude is insensitive to halo flattening ($q = 0.8$ – 1.0) in the presence of a strong tidal perturber (Section 3.5.3).

(vii) **Comparison with observations.** The fiducial model (peak tilt 1.70° , outer-disk tilt 1.34°) lies at the lower end of the observed MW stellar warp (2° – 5° ; Chen et al. 2019; Skowron et al. 2019). The deficit is likely attributable to our neglect of the dynamical halo response, disk truncation at $R = 18$ kpc, and the absence of a second perturber (Section 4.1).

The tilted-ring framework, while simplified, provides a physically transparent and computationally efficient tool for understanding the parametric dependencies of tidal warp formation. Future work should incorporate a live dark matter halo to quantify the amplification of the tidal response, extend the disk to larger radii, and include gas dynamics to capture the differential stellar/gaseous warp response.

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DATA AVAILABILITY

The simulation code and all output data are publicly available at <https://github.com/XiangJinyu/stellar-warp-sim>.

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